Trial Autopsy: how the sponsor technology is used

How Zetaris, Meterless and NVIDIA are used, and what is distinctive about each. Every claim points to the code and to a way to check it. The numbers come from the 17 recorded analyses (14 blind backtests and 3 examples, run on 22 and 23 October 2026).

Team CurrentSky · Stanislav Gatin (solo) Repository: `docs/SPONSOR_TECHNOLOGY.md`, `metadata.json`

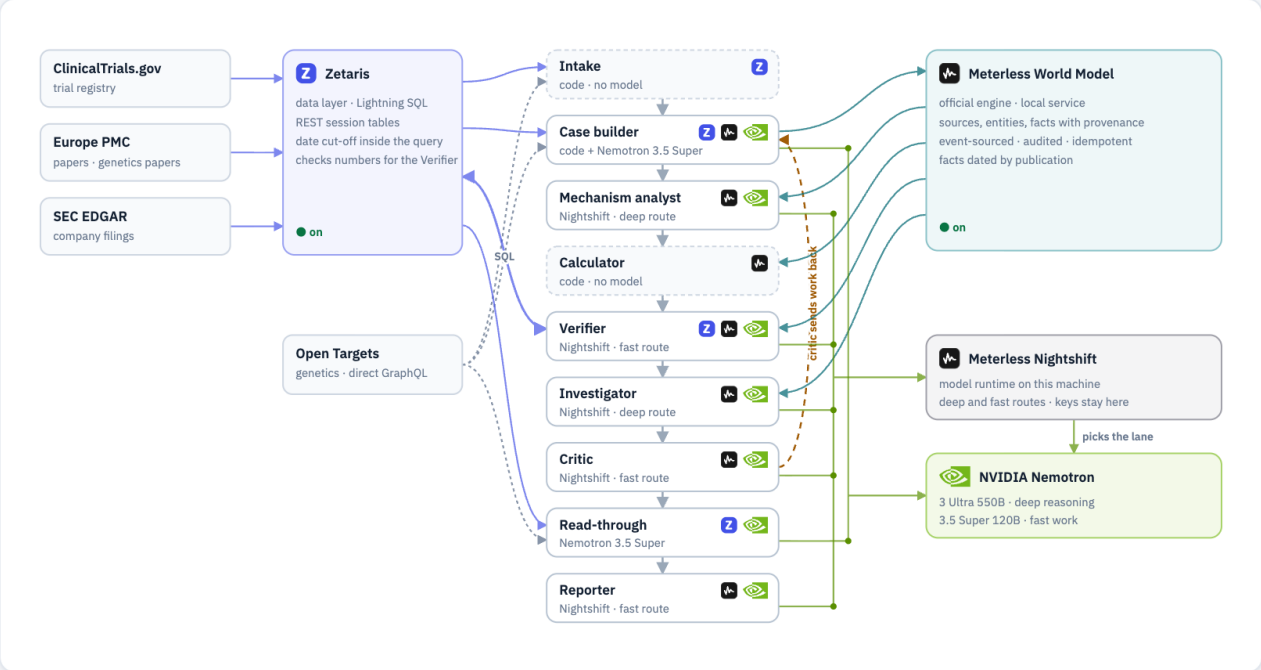
At a glance

Sponsor	Role in the product	Code

Zetaris	Data layer: the trial registry, the literature and company filings are read through Zetaris with SQL, and the date cut-off runs inside the query. The Verifier has Nemotron write the SQL that checks every number the calculator uses against its source.	`app/tools/zetaris.py`
		`app/tools/http.py`
		`app/tools/europepmc.py`
		`app/agents/verifier.py`

Meterless	Two roles. **Power:** Meterless Nightshift is the primary model provider. **Build:** the official World Model engine stores every source, entity and fact with its provenance.	`app/llm/meterless_runtime.py`
		`app/llm/generator.py`
		`services/meterless/`
		`app/engines/world_model.py`

NVIDIA	Every AI step runs on NVIDIA Nemotron (3 Ultra and 3.5 Super).	`app/config.py` `app/llm/generator.py`



The system map from the app's Settings page: data sources and Zetaris on the left, the nine steps in the middle, the Meterless World Model, Meterless Nightshift and NVIDIA Nemotron on the right. Sponsor logos mark each step that uses that sponsor's technology.

Zetaris: the data layer

What it does. Trial Autopsy reads ClinicalTrials.gov, Europe PMC and SEC EDGAR through a Zetaris instance instead of calling those APIs directly. Each request is registered as a Zetaris REST session table and read back with Lightning SQL over the Zetaris REST API:

```
CREATE REST SESSION TABLE ta_<id> REQUEST (endpoint "<source URL>", method "GET", ...)
  HEADER (user-agent "...") BODY ()
SELECT to_json(r) FROM session.ta_<id> LATERAL VIEW explode(resultList.result) AS r
  WHERE r.firstPublicationDate <= '<as_of>'
```

Why it matters here. A blind analysis may only see documents published before the trial's result. For the literature, Zetaris applies that cut-off inside the SQL query, so a paper from after the cut-off is filtered out by the data layer before any agent sees it.

Resilience. If Zetaris is unreachable, the read falls back to the direct request and the run trace records a warning. Open Targets (a GraphQL POST API) is read directly.

The Verifier: Nemotron writes SQL, Zetaris checks the numbers

After the calculator has run, the Verifier takes each number it used (a biomarker change, planned events, follow-up) and checks it against its source through Zetaris:

1. The registry record and the cited papers are registered as Zetaris REST session tables (aliases `registry` and `papers`).
2. Nemotron writes one Lightning SQL query per number, for example:

   ```
   SELECT regexp_extract(r.abstractText, '([^.]*993[^.]*)', 1) FROM papers
     LATERAL VIEW explode(resultList.result) AS r WHERE r.pmid = '40185318'
   ```

3. Code accepts only one SELECT over the known tables, with no comments or other statements, and adds a row limit. In a blind backtest it also refuses fields known only after the readout: posted results, status, actual enrolment.
4. Zetaris runs the query. If Zetaris returns an error, Nemotron gets one chance to fix the SQL.
5. Nemotron judges each number from the rows that came back: confirmed, not found or contradicts. Code adds one rule: a number with no rows is never confirmed.

A contradiction goes to the critic as a serious issue, and the critic can send the work back (up to two rounds). Without Zetaris, the report says its numbers were not independently verified and lowers its confidence one level.

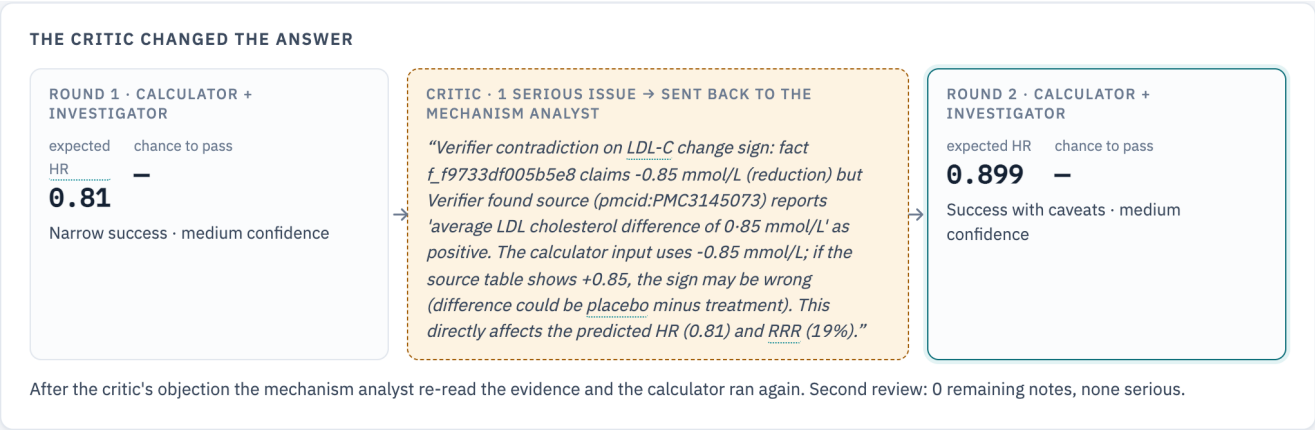
Independent source check: Verifier, SQL run in Zetaris

Nemotron wrote one Lightning SQL query per number; Zetaris ran it over the trial's registry record and the cited papers. 6 confirmed, 0 not found, 1 contradicted.

NUMBER	USED AS	RESULT	FOUND IN THE SOURCE	QUERY
`-58.9 %` `f_6b128b9c595750 · pmid:36529659`	Lp(a) change used by the calculator	Checked in Zetaris ✓	*0008), -58.9% (p=0*	`SELECT regexp_extract(r.abstractText, '([^.]*58\.9[^.]*)', 1) AS sentence FROM papers LATERAL VIEW explode(resultList.result) AS r WHERE r.pmid = '36529659' LIMIT 5`
`70 mg/dL` `f_71b61a5fd20704 · pmid:40185318`	Lp(a) change used by the calculator	Checked in Zetaris ✓	*elevated Lp(a) levels of ≥ 70 mg/dL (approximately 149 nmol/L)*	`SELECT regexp_extract(r.abstractText, '([^.]*70[^.]*mg/dL[^.]*)', 1) AS sentence FROM papers LATERAL VIEW explode(resultList.result) AS r WHERE r.pmid = '40185318' LIMIT 5`
`-19 %` `f_4b4e03d46ecbee · pmid:35300814`	LDL-C change used by the calculator	Checked in Zetaris ✓	*Pelacarsen resulted in mean percent/absolute changes of -2% to -19%/-0*	`SELECT regexp_extract(r.abstractText, '([^.]*19%[^.]*)', 1) AS sentence FROM papers LATERAL VIEW explode(resultList.result) AS r WHERE r.pmid = '35300814' LIMIT 5`
`-8 mg/dL` `f_06dddc238f9482 · pmid:35300814`	LDL-C change used by the calculator	Checked in Zetaris ✓	*7 to -8.0 mg/dL (P = 0*	`SELECT regexp_extract(r.abstractText, '([^.]*8\.0[^.]*mg/dL[^.]*)', 1) AS sentence FROM papers LATERAL`

The *Numbers used* tab of the pelacarsen report: the number, the verdict, the sentence Zetaris found and the SQL Nemotron wrote.

Across the 17 analyses the Verifier checked 85 numbers: 50 confirmed, 31 not found in the abstract or registry, 4 contradictions. Each contradiction reached the critic. In IMPROVE-IT the source reported the LDL-C difference with the opposite sign to the input; the critic sent the work back, and the second round's expected hazard ratio, 0.899, matched the answer key's 0.898. In pelacarsen the 2.5-year follow-up was a minimum, not the median, and the second round used 3.25 years.



IMPROVE-IT: the Verifier's contradiction becomes the critic's serious issue, and the second round changes the answer.

HOW TO CHECK

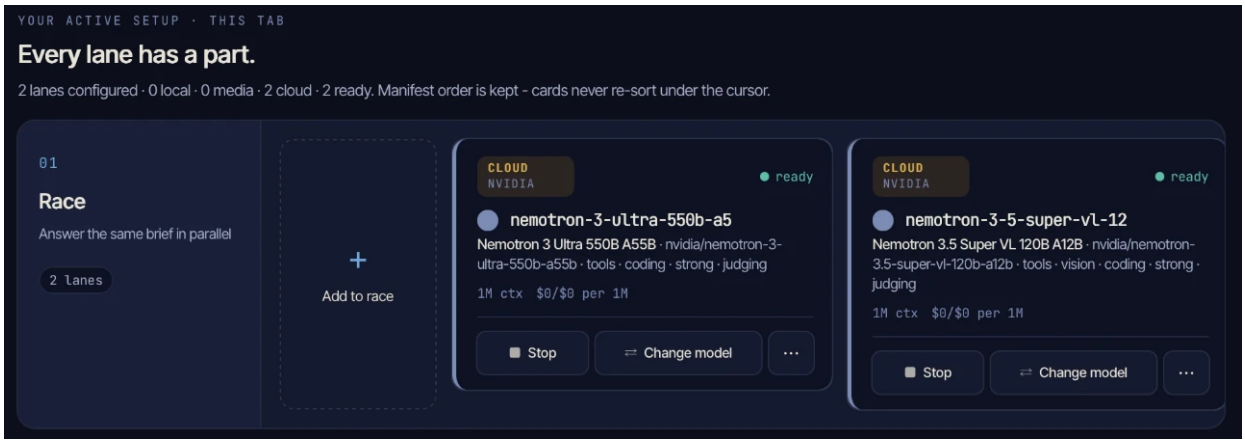
- Settings page → Zetaris card → *Test connection*; `scripts/smoke_test.py` runs `SELECT 1` over the REST API.
- A live run's step log shows `read via Zetaris`, `register_tables`, `write_sql` and `query_zetaris` lines.
- The report footer says `data layer: Zetaris (N source reads via Lightning SQL)`; the *Numbers used* tab shows every query and verdict.
- `tests/test_verifier.py` covers the SQL guard, the cut-off rule, the run without Zetaris and a contradiction reaching the critic.

Meterless Nightshift: the model runtime (Power)

What it does. Nightshift is the app's primary AI provider, following the Meterless Runtime provider contract (`app/llm/meterless_runtime.py`):

- it calls Nightshift's OpenAI-compatible endpoint (default `http://127.0.0.1:8088/v1`) with the runtime token from Nightshift's Console;
- every request carries the header `meterless-app: trial-autopsy`, and chat requests also carry `meterless-scope: user`;
- each step asks for a routing mode, chosen per step on the Settings page from Nightshift's `GET /v1/models` list (`meterless/deep`, `meterless/fast`, `meterless/auto`); Nightshift picks the Nemotron lane that answers;
- Settings has Auto Detect (ports 8088, 8090, 8787, 11434), paste of a connect link, and a connection test that reads `/v1/models` and sends one request.

The provider keys stay in Nightshift. Each response's `model` field names the lane that answered, and the trace records it. Answers are stored under the model that actually answered, so one lane's answer is never replayed as another model's.



Nightshift's lanes: Nemotron 3 Ultra and Nemotron 3.5 Super VL on its NVIDIA provider, both ready.

Through Nightshift in the recorded runs. Nightshift served the five reasoning steps: the mechanism analyst, the Verifier, the investigator, the critic and the reporter. In the 16 live runs all 132 of those calls went through Nightshift and none needed the direct fallback; Nightshift answered 131 of them from Nemotron 3 Ultra. A full analysis took a median of 10 minutes (5 to 33). Per-run table: `docs/evidence/nightshift_runs.md`.

What we learned running a pipeline through Nightshift

Nightshift is built first for chat. A pipeline of strict steps needs a few settings, and we measured each:

What we found	Effect on the pipeline	How we handle it
Build 0.92.0 hid NVIDIA's models in the lane picker (`nvidia` missing from `byokProviders`)	Nemotron could not be added to a lane	Reported with the one-line fix (Meterless/Meterless#4); fixed in 0.92.1 the same night
Tier-split, on by default, adds a planner phase before the request	Strict extraction rules were diluted: one document gave 9 facts directly and 0 through Nightshift	Tier-split off in *Setup → Preferences*
The reasoning switch (`chat_template_kwargs`) is not passed on	With reasoning on, extraction found 0 to 3 facts where reasoning off found 9	Extraction and read-through call NVIDIA's API directly with reasoning off
For an identified app, the mode bound in the Console decides the routing	One routing mode for the whole app	The app is bound to `fast`; Nightshift picks the lane
A cloud request with no first token after 45 s is cut	Long reasoning prompts on a busy Nemotron 3 Ultra were cut	Streaming, a watchdog, retries, and a direct NVIDIA call after two cuts (needed 0 times in the final runs)
An answer cache keyed on the last 4,000 characters of the prompt	One run got the previous run's report back; a repeated connection test was answered from the cache	Every request carries a unique reference, so the cache cannot replay a stale answer

HOW TO CHECK

- Settings page → Meterless card → *Auto Detect*, then *Test connection*: the routing modes Nightshift serves and the lane that answered.
- `scripts/smoke_test.py`: "Nemotron replies through Meterless Nightshift" passes only if a Nemotron model answered.
- In a run's step log, every reasoning step shows `meterless:nvidia/nemotron-....`

Meterless World Model: the evidence store (Build)

What it does. The official Meterless World Model reference engine (from github.com/Meterless/Meterless, unchanged, Apache-2.0) runs on the machine as a local service in `services/meterless/`. `run.py` starts it, and the Docker image includes it. Every source, entity and fact the agents produce is written to it, one namespace per run, with provenance: the step that found it; the run; the source URL; the source's publication date, used as the observation time.

Facts whose quote failed the word-for-word check are stored too, marked unverified with low confidence; only verified facts reach the calculator and the report. The engine keeps an event-sourced, audited, idempotent world state. `app/engines/world_model.py` implements the same three services the Meterless engine guides describe (WorldStore, IngestPipeline, ContextBuilder), and the analyst, the calculator, the Verifier and the investigator build their context from them. Across the 17 analyses the engine stored 374 entities, 1,331 facts, 445 links and 3,373 audit entries.

HOW TO CHECK

- The official conformance suite passes 8 of 8 checks:
  `npm –prefix services/meterless run conformance`
- Settings page → Meterless World Model → *Test connection*; `scripts/smoke_test.py` checks the service.
- Each run's step log ends with a `world_model_sync` line, for pelacarsen: 18 entities · 58 facts · 19 links · 147 audit entries.

NVIDIA Nemotron: the models

What it does. Every AI step runs on NVIDIA Nemotron.

Step	Model in the recorded runs	Route	Reasoning
Mechanism analyst: picks the numbers the calculator uses	Nemotron 3 Ultra (550B)	Nightshift, deep	on
Verifier: writes the SQL that checks each number in Zetaris	Nemotron 3 Ultra (550B)	Nightshift, fast	on
Investigator: weighs every possible cause	Nemotron 3 Ultra (550B)	Nightshift, deep	on
Critic: checks the work and can send it back	Nemotron 3 Ultra (550B)	Nightshift, fast	on
Reporter	Nemotron 3 Ultra (550B)	Nightshift, fast	on*
Extraction: copies numbers with verbatim quotes	Nemotron 3.5 Super (120B)	NVIDIA API	off
Read-through: rival drugs on the same target	Nemotron 3.5 Super (120B)	NVIDIA API	off

*Requested off; Nightshift does not pass the switch on. Nightshift picks the lane for each request; with the app bound to `fast`, it served the reasoning steps almost entirely from Ultra.

Built for open models

- The critic has its own instructions and reviews the analyst's work, and the Verifier checks the numbers against the sources, so the steps check each other.
- Calls are paced under the API's rate limit; malformed JSON is repaired by the same model.

Code does the checks the model could get wrong

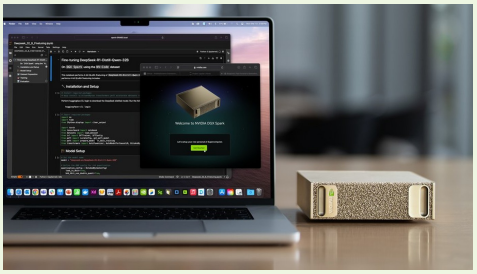
Numbers come from verified quotes. Code then sets the direction of a change from the source's own words (reading the start of the sentence when a quote begins mid-sentence), converts units, and fills planned events or a baseline from cited facts. The run trace records each such step.

Measured

The same 14 finished trials, replayed blind with only the documents published before each result, scored against the explanations experts published later (`eval/results/backtest_2026-10-23.md`):

The same 14 trials, blind	**NVIDIA Nemotron through Meterless Nightshift**
Success or failure called right, where the trial could be sized	**9 / 9**
Experts' cause as the main cause	**10 / 14**
Experts' cause among those named	**11 / 14**
Expected hazard ratio within 0.03 of the answer key	**10 / 13**

Every recorded run replays offline with the same outcome, expected hazard ratio, power and fact count as the live run.

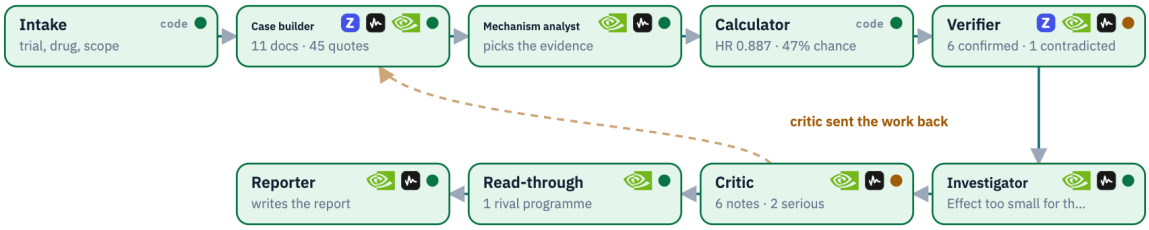


NVIDIA DGX Spark next to a Mac. Image: NVIDIA

Why open models. A drug company cannot send unpublished trial data to a cloud AI it does not control. Because every step runs on open Nemotron models, the same analysis can run on the company's own hardware, for example an NVIDIA DGX Spark, which Nightshift supports. In Trial Autopsy the switch is one setting, the model address; the agents, prompts and checks stay the same. We ran on NVIDIA's API and on Nightshift on a Mac; we have not run the system on a DGX Spark.

HOW TO CHECK

- Settings page: every AI step shows the NVIDIA logo and its Nemotron route or model.
- Each report's agent graph and *What this analysis used* panel show the model each step actually used.
- `eval/results/` holds the scores; `data/library/traces/` holds every recorded run's step log.



The agent graph of the pelacarsen run: each step shows the technology it ran on.

What is distinctive

Zetaris enforces the time machine.
A blind analysis cannot leak future information: SQL removes later papers before any agent sees them.

Zetaris checks the numbers.
Nemotron writes the SQL, code guards it, Zetaris runs it, and a contradiction goes to the critic. In IMPROVE-IT that loop moved the expected hazard ratio from 0.81 to 0.899, against 0.898 in the answer key.

Meterless in both required roles.
Every reasoning call in the recorded runs went through Nightshift; the World Model engine runs unchanged and passes its conformance suite. We reported a Nightshift bug with its fix (Meterless/Meterless#4), fixed the same night, and documented the settings a strict pipeline needs.

Open models, measured.
The whole pipeline runs on open NVIDIA models and is scored on blind backtests against expert explanations, with every recorded run reproducible offline.

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